
An In-Silico Instrument for Pre-Validating Corpus-Bounded Instruction: Reference-Policy Transfer Measurement with Bidirectional Corpus-Gap and Leakage Diagnosis

Anonymous authors
Paper under double-blind review

Abstract

1 When a model answers correctly with a corpus in context, did it *need* the corpus, or
2 did it already know the answer? We present an *in-silico measurement instrument*
3 that answers this—and the dual question of whether a corpus is *sufficient*—as a
4 behavioral task outcome, before any human-subject study. The central quantity is
5 *corpus necessity*: ablate a corpus item and measure the change in a task outcome
6 scored against a reference. Our headline contribution is that we make necessity
7 *non-confounded* by calibrating it against a *known parametric baseline*: we *con-*
8 *struct* the ground truth by teaching a fact into a model’s weights and showing
9 necessity falls as a *dose-response*. We demonstrate this in **two domains with in-**
10 **dependent objectives**, one of which uses *no simulator at all*: (i) a control-regret
11 simulator (maritime collision avoidance), where teaching a hidden charted haz-
12 ard into the weights collapses necessity from a full-barrier swing to ≈ 0 , with
13 the LoRA-scale and training-checkpoint gradients agreeing; and (ii) a simulator-
14 free *fictional-fact question answering* task scored by an answer checker, where
15 teaching invented facts into a 7B model drives necessity *smoothly and monoton-*
16 *ically from 1.00 to 0.03*. This calibrated necessity measure is the basis of two
17 products: a *corpus diagnosis* that ranks each item by necessity, localizes which
18 item is missing from task failures, and—crucially—splits an unrelieved-upon item
19 into *redundant* (the model already knows it) versus *unusable* (the model fails even
20 with it present); and a *corpus-sufficiency* verdict that surfaces *false sufficiency*—a
21 corpus that only appears sufficient because the model is coasting on priors and is
22 therefore fragile to distribution shift. Unlike faithfulness (RAGAS) and context-
23 attribution (ContextCite), which measure whether context *supported* or *influenced*
24 an output, we measure whether the model *needed* it, against a weight-level base-
25 line they lack. The method is *closed-model compatible* (API access only) and
26 operates *per item*. As breadth we report: a *learner ladder* that reads corpus contri-
27 bution beyond mere existence—decision *quality* (avoiding well, not just avoiding)
28 and *reasoning depth* (using a local rule that overrides a reflex, not just looking one
29 up); cross-model discrimination across current LLMs; a second control domain
30 (roadside tire change); KC→single-metric identifiability and estimator-recovery
31 calibration; and a machine-unlearning arm that yields a *says* $\neq$ *does* dissociation
32 (gentle unlearning changes what a model *says* but not what the instrument sees it
33 *do*) as supporting weight-level evidence. All results are **simulation-based mecha-**
34 **nism evidence, not human-subject external validity**; we scope the contribution
35 accordingly and pre-register the human study as future work.

1 Introduction

Generic retrieval-augmented (RAG) assistants optimize for semantic relevance: give a plausible answer to whatever is asked. That objective is wrong for training in safety-critical procedural domains, where a semantically accurate answer can still be instructionally wrong—delivered at the wrong complexity, skipping a prerequisite, or worse, confidently fabricated. A growing body of work therefore builds *corpus-bounded* tutors that abstain outside a curated corpus. But a designer who has built such a system faces a prior question: *does the mechanism work?* Does the tutor actually rely on its corpus, and does the corpus actually cover the domain? Answering this by human-subject study is slow and expensive, and a null result there conflates many causes.

We argue that the *mechanism* of a corpus-bounded method can be falsified cheaply and in silico, and we build the instrument to do it (Figure 1). Our contribution is a *methods/measurement* one, not a new tutor and not a new collision-avoidance technique. Concretely:

The central storyline (Sections 4–6):

- **Necessity, calibrated by construction.** We measure *corpus necessity*—the change in a task outcome when a corpus item is ablated—and make it non-confounded by *constructing* the parametric baseline: teach a fact into the weights and necessity falls as a *dose-response*. We show this in **two domains with independent objectives**, one *simulator-free*: a control-regret simulator (teaching a hidden hazard collapses necessity to ≈ 0 ; LoRA-scale and checkpoint gradients agree) and a simulator-free fictional-fact QA task scored by an answer checker (teaching invented facts drives necessity smoothly $1.00 \rightarrow 0.03$ on a 7B model). This is a known-groups validation of the measure.
- **The corpus diagnosis** ranks each item by necessity, localizes a missing item from task failures, and splits an unrelayed-upon item into *redundant* (already known) versus *unusable* (present but the model fails with it)—distinct fixes (prune vs. fix the model).
- **Corpus sufficiency**, the product face of the audit, surfaces *false sufficiency*: a corpus that only looks sufficient because the model coasts on priors (necessity ≈ 0 everywhere), a fragility ordinary RAG metrics report as all-green.

Additional breadth (supporting the above, not the headline): a *transfer instrument* that scores a learner by its *regret against a reference-policy solver* with KC→single-metric identifiability (Section 3); a *learner ladder* that shows the same instrument reads corpus contribution beyond existence—decision *quality* and *reasoning depth* (Section 5); a second control domain and estimator-recovery calibration; cross-model discrimination on current LLMs (Section 4); and a machine-unlearning arm that yields a *says $\neq$ does* dissociation as secondary weight-level evidence (Section 6).

Our contribution is a *methods/measurement* one, not a new tutor. Every *component* we build on is prior art; the *calibrated necessity measure* and its *corpus-value/sufficiency audit* are the open ground (Section 2). We are explicit throughout that our evidence is simulation-based: it tests whether the machinery behaves as designed, not whether humans learn more.

Organization: four case studies. The empirical work is a register of experiments grouped into four case studies (Table 1), which we reference throughout as CS $X.Y$. **CS1** builds and stress-tests the necessity instrument on the COLREG core (the corpus diagnosis and sufficiency products, Section 4). **CS2** climbs a *learner ladder*—blind, naive, implementer, reasoner—to show the same instrument reads not just *existence* (did you avoid at all) but *quality* (did you avoid well) and *reasoning depth* (do you generalize a local rule), Section 5. **CS3** enriches for reach and generality (a simulator-free second domain, a discrete procedural domain, real charted hazards). **CS4** validates the instrument itself (construct validity, reweighting sensitivity, the unlearning null, an offline reproduce runner). The headline necessity claim is validated by construction in two domains with independent objectives: CS1.7 (control-regret, the large-effect known-groups endpoints) and CS3.2 (simulator-free fact-QA, the graded monotone curve).

2 Related work and positioning

Each strand below is established prior art; we reuse it as infrastructure and mark where our delta departs from it.

Case study	Where	Key experiments (CSX.Y)
CS1 necessity instrument (COLREG core)	§4	1.1 reference-learner recovery; 1.2/1.3 cross-model leakage & hazard discrimination; 1.4 geometric hazard suite (necessity as a fraction); 1.5 redundant-vs-unusable; 1.6 corpus-value audit/localization; 1.7 hazard dose-response (§6.1); 1.8 sufficiency & false-sufficiency
CS2 blind→expert ladder (quality→reasoning)	§5	2.1 decision-quality band (the middle band); 2.2 reason-vs-implement (generalization axis)
CS3 enrichments (reach & generality)	§4, 6.1	3.1 real-hazard screen; 3.2 fact-QA necessity (no simulator); 3.3 discrete procedural domain (§3); 3.4 probe-suite breadth
CS4 validations (does the instrument hold up?)	§3, 6.2	4.1 construct validity; 4.2 reweighting sensitivity ($\tau=1.00$); 4.3 unlearning null (says $\neq$ does); 4.4 offline reproduce runner

Table 1: The experiment register, grouped into four case studies and referenced as CSX.Y throughout. The sections cut across the case studies (the instrument’s forward validation, CS4/CS3.3, is Section 3; its backward necessity use, CS1, is Section 4; the weight-level calibration, CS1.7/CS3.2/CS4.3, is Section 6); this map is the reading guide.

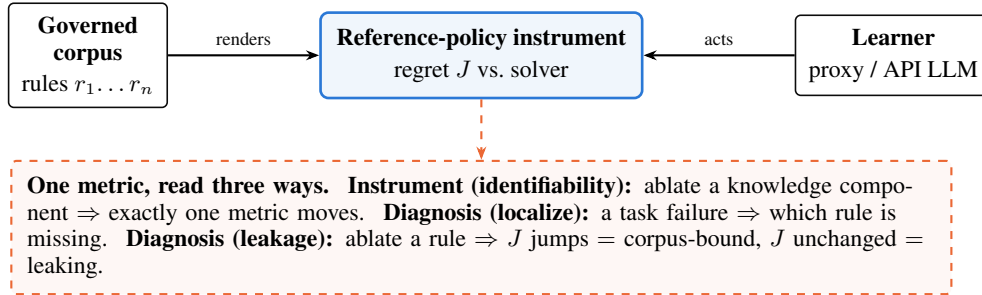


Figure 1: The instrument, used bidirectionally. A governed corpus and a learner both feed one reference-policy simulator that scores the learner’s *regret* J against an automatic solver. The same metric is then read three ways—to verify KC→single-metric identifiability (the *instrument*), to localize a missing rule from task failures, and to detect parametric leakage by rule ablation (the *diagnosis*)—needing only API access to the learner. The diagnosis is the instrument run backwards, not a separate tool.

87 **Learner modeling and instructional theory.** Knowledge tracing [Corbett and Anderson, 1994],
88 the Elo rating system for adaptive systems [Pelánek, 2016], the Knowledge-Learning-Instruction
89 framework [Koedinger et al., 2012], open learner models [Bull and Kay, 2007], and the Zone of
90 Proximal Development [Vygotsky, 1978] are all established. We reuse them as infrastructure, not
91 as contributions.

92 **Corpus-bounded tutoring, leakage, and faithfulness.** RAG tutors that abstain outside a corpus
93 are standard engineering. A large line asks whether a model uses provided context or parametric
94 memory: faithfulness metrics score whether an answer is grounded in context [Es et al., 2023]; con-
95 text attribution ablates context sources and measures the change in output probability to attribute
96 a generation to them [Cohen-Wang et al., 2024, Or, 2026]; knowledge-conflict work studies which
97 side wins when context contradicts memory [Xu et al., 2024, Zhang et al., 2025]; leakage-free bench-
98 marking filters items answerable with no context [Liu et al., 2026a]; and intermediate-step scoring
99 looks beyond the final answer [Jain and Vedam, 2026]. All of these measure whether context *in-*
100 *fluenced* the output, judged on *answer text*, per answer or system-wide. We differ on three axes:
101 (a) we score a *behavioral task outcome* against a reference-optimal policy, not answer text; (b) we
102 attribute *per corpus rule* and localize which rule is missing or wrong, not per token or system-wide;
103 and (c) we measure not merely *influence* but *necessity*—calibrated by manipulating the parametric

baseline directly (weight-level construction and unlearning, Section 6), so a fact the model already knew is not mistaken for one it relied on. The gap is structural: a faithfulness score is a function of the *(question, context, answer)* triple computed with the corpus *present*, whereas necessity is a *contrast*— $\text{acc}(\text{with corpus}) - \text{acc}(\text{without corpus})$ —and RAGAS never runs the without-corpus arm, so it is *structurally blind* to the quantity: a model that knows a fact and one that does not both score high with the corpus present, and the redundant case is reported as sufficient—the false-sufficiency failure our measure is built to catch. (Our released harness computes both side by side.) **ContextCite** [Cohen-Wang et al., 2024] is the closest relative: because it ablates context and measures the change in the response’s log-probability against the model’s own baseline, it *partially* tracks necessity—but it requires open weights and per-token log-probabilities (so it cannot run on the closed API models our necessity test does), attributes text-influence rather than a task outcome, carries no weight-level known-groups calibration, and yields no *redundant/unusable* split or sufficiency verdict. That join is the *Corpus Diagnosis*.

Objective competency instruments. Deviation-from-optimal scoring appears in flight [Zinn et al., 2023] and construct-validity-by-novice/expert-separation in surgery [Van Dongen et al., 2007]; control-theoretic notation has been applied to knowledge dynamics [Anonymous, 2024b]; regret is used in RL transfer [Tuynman and Örtner, 2022]; maritime training analytics regress simulator measures onto instructor ratings [Lall and Liu, 2025]. None uses a *reference solver’s regret* as the *learner* transfer outcome with $\text{KC} \rightarrow \text{single-metric}$ identifiability. That is the *Transfer Instrument*. We borrow mature maritime-autonomy tooling—velocity obstacles [Fiorini and Shiller, 1998] and simulation-based MPC [Johansen et al., 2016]—purely as reference solvers, never as a new collision-avoidance method [Anonymous, 2024a, International Maritime Organization, 1972].

Proxy learners. Simulated students have long authored and evaluated tutors; LLM student agents now pilot assessment items [Lu and Wang, 2024] and simulate learners for tutoring-system development [Gao et al., 2025], and controlled variants suppress knowledge by prompting [Apartsin et al., 2026], interpolate parametric competence [Liu et al., 2026b], or unlearn a model into a novice [Song et al., 2026]. Pre-deployment simulated-learner screens have very recently appeared [Fan et al., 2026, Lee et al., 2026]. We therefore *demote* the proxy-learner idea to an application: our narrow residual claim is running *two complementary proxy classes* (parametric competence *and* unlearned LLM) scored on the *same regret-to-reference instrument* the human trial would use—which none of these do.

3 The Transfer Instrument

This section defines and verifies the instrument’s *forward* use—measuring transfer and skill. The *backward* use that measures corpus necessity, the paper’s central contribution, follows in Section 4; the construct-validity, identifiability, and robustness checks below are the support that licenses it.

Task and metric. The worked domain is COLREG collision avoidance: a power-driven vessel must choose course and speed to resolve an encounter. We implement a kinematic simulator with an elliptical ship domain and a Collision Risk Index, and two reference solvers—velocity obstacles [Fiorini and Shiller, 1998] and simulation-based MPC [Johansen et al., 2016]. For a learner policy π on scenario set S we define the transfer outcome as regret against a reference policy π^* ,

$$\mathcal{R}(\pi) = \frac{1}{|S|} \sum_{s \in S} [J(\pi, s) - J(\pi^*, s)], \quad (1)$$

where J is the per-encounter objective (a weighted sum of a safety-margin barrier, COLREG compliance penalty, and route deviation; lower is better) and π^* is the reference maneuver. Unlike item correctness, $\mathcal{R}$ is continuous, objective, and instructor-free.

On “reference,” not “optimal.” We deliberately avoid claiming a global optimum. VO and SB-MPC are strong, widely-used near-optimal solvers, but neither is proven globally optimal for the full nonconvex encounter; calling their output “optimal” would overclaim. The instrument does not need a true optimum—only a reference that construct validity shows separates naive from expert behavior (below). Regret is thus measured against a fixed, reproducible reference policy, and we report which solver is used.

Policy	mean J	median	min	max	cleared
naive (hold course)	1765.44	1988.08	1.08	1995.18	11%
VO reference	0.70	0.58	0.33	1.27	100%
SB-MPC reference	0.01	0.01	0.00	0.03	100%

Table 2: Construct validity and grading across nine varied encounters. The objective J separates the three competence levels cleanly and takes a continuous range of values (20 distinct across all cells, $[0.00, 1995.18]$)—it is a graded transfer metric, not a near-binary check. `npm run colreg:construct`.

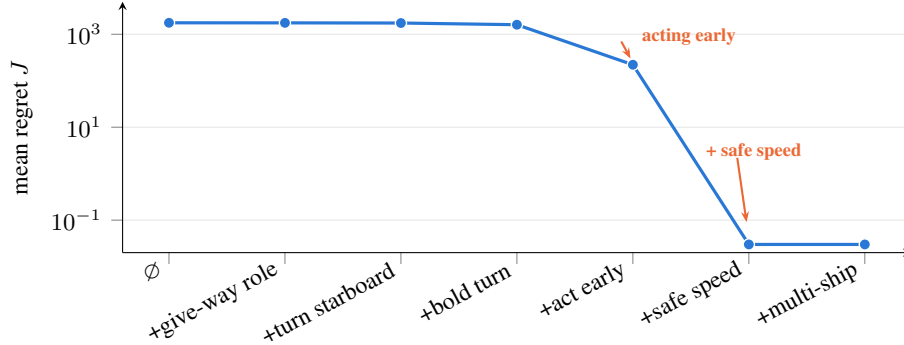


Figure 2: Competence→performance gradient on the nine-encounter instrument. Mean regret J (log scale) as the six knowledge components are acquired in curriculum order (left to right); each point is a competence stage, from the empty learner ($\emptyset$) to full competence. J spans nearly five orders of magnitude and collapses as the two domain-clearing competencies—acting *early* and at a *safe speed*—come online, the graded behavior Table 2 reports numerically. `npm run colreg:construct`.

153 **Construct validity and metric grading.** The instrument must separate good from bad policies
154 before any learning claim, and—if it is to be a transfer measure rather than a pass/fail check—
155 the metric must *grade*, not collapse to two values. Across a varied set of nine encounters (head-on,
156 starboard/port crossing, overtaking, and restricted-visibility geometries), three policies of increasing
157 competence separate cleanly and continuously on J (Table 2): a naive hold-course policy (mean $J \approx$
158 1765, clearing only 11% of domains), a velocity-obstacle reference ($J \approx 0.70$, all clear), and SB-
159 MPC ($J \approx 0.01$). The objective takes 20 distinct values in $[0.00, 1995.18]$ over the policy $\times$ scenario
160 cells—a graded metric, with the safe regime (VO/SB-MPC) itself spanning a continuous 0.00–1.27
161 range. On the discrete procedural domain the same logic holds: an expert policy scores $J = 0$
162 against a reckless $J = 200$ with the safety violation caught. This is the novice/expert-separation
163 logic of Van Dongen et al. [2007], applied to a decision/control task rather than manual skill.

164 **KC→single-metric identifiability.** We design each knowledge component to govern exactly
165 one metric, so ablating a KC degrades that metric and no other. On the procedural domain
166 this is *exact*: each ablated competence degrades only its governed metric, and a monotone
167 competence→performance gradient ($J : 204 \rightarrow 0$) follows as competence is acquired in curricu-
168 lum order; on the COLREG instrument the same gradient spans nearly five orders of magnitude
169 (Figure 2). Because the property holds on both a continuous-control and a discrete-procedural in-
170 strument, it is a general measurement-design property, not a COLREG artifact.

171 **Is identifiability tautological?** A fair objection is that the mapping is a design choice, so “ablate
172 the KC, that metric moves” is true by construction. But the *cross-effect* claim can fail: designing
173 the metrics does not guarantee that ablating one KC leaves the *others* unchanged—if the objective’s
174 terms are entangled, ablation bleeds across metrics. Identifiability is the empirically-checked *ab-*
175 *sence* of that bleed, and it is what makes the Corpus Diagnosis (Section 4) actionable (a task failure
176 attributes to a specific rule). We thus treat it as a design requirement to be verified per instrument. On
177 the procedural domain the verification is clean (Table 3): each ablation degrades only its governed
178 metric, maximum off-diagonal change 0.0000—no cross-bleed.

ablated KC ↓ / metric →	safety	core	sequencing	finishing
safety	−1.00	0.00	0.00	0.00
core steps	0.00	−1.00	0.00	0.00
sequencing	0.00	0.00	−0.22	0.00
finishing	0.00	0.00	0.00	−1.00

Table 3: KC→single-metric identifiability (procedural domain). Change in each metric after ablating each knowledge component. The diagonal (bold) is the intended governed metric; every off-diagonal cell is 0.00 (max $|\Delta| = 0.0000$)—ablation does not bleed across metrics. `npm run proc:identifiability`.

Estimator	Recovery of known ground truth	Calibration (ECE)
Elo	static ability, mean abs. error 0.046	0.023
BKT	latent state $P(\hat{L}) = 0.997$ (learned) vs. 0.175 (not)	0.004

Table 4: Estimator-recovery and calibration on synthetic data with a known ground truth. The measurement, not just the metric, is validated.

Robustness to weighting choices. The objective embeds weighting choices—the elliptical ship-domain radii and the margin/compliance/deviation weights—and a fair worry is that the competence ranking is an artifact of those choices. We falsify this with a sensitivity analysis (`npm run colreg:sensitivity`) that holds every policy’s *behavior* fixed and re-scores under 232 perturbations: a one-at-a-time ± 25 –50% sweep of each parameter plus 200 joint samples with every parameter independently scaled in $[0.6, 1.4]$. Across the whole ensemble, both orderings are preserved exactly—the naive policy stays worst and the six-component competence gradient stays monotone in J (Kendall $\tau = 1.00$ on every perturbation, invariant rate 100%). The CRI parameters are excluded because CRI is a reported diagnostic, not a term in the scored objective, so the ranking is invariant to them by construction. The protocol’s conclusions are thus a property of the competence structure, not of the weight calibration.

Estimator recovery and calibration. Feeding the instrument’s outcomes into a generic learner-agent estimator, we recover known ground truth on synthetic data (Table 4). This turns “we have a metric” into “the measurement is valid.”

4 The Corpus Diagnosis: the same instrument, read backwards

The Corpus Diagnosis is not a second tool. It is the Transfer Instrument of Section 3 run in reverse: the *same* objective metric answers two opposite questions. This backward reading—*does the learner rely on a given corpus item?*—is the paper’s central contribution; the forward construct validity of Section 3 is what licenses it.

(i) Corpus-gap localization. When the corpus is missing or wrong on a rule, the learner’s failures on the governed metric have a signature that attributes the failure to a *specific* rule/node. We compute, per rule, an ablation-delta—the rise in the governed penalty when that rule is removed—and a localization that names the governed component. A large delta plus a matching localization means the learner *relied on* that corpus rule.

(ii) Parametric leakage detection (necessity). The mirror image: ablate a rule from the corpus; if the metric it governs does *not* degrade, the model is answering from pretrained priors, not the corpus. We combine the ablation-delta with a *counterfactual* test (replace “alter to starboard” with “alter to port”; does the learner follow the altered rule?) and a *closed-book* contamination baseline (no corpus supplied). The verdict—CORPUS-BOUND, LEAKING, or INCONCLUSIVE—is read from the same metric, by a majority vote over the three signals so that a clear ablation-and-contamination pattern is not vetoed by a single dissenting sub-test. Section 6.1 calibrates this necessity against a known parametric baseline by construction. Because the bounded compliance penalty is a *term inside* the objective J , this necessity is read off the graded regret instrument itself—not a decoupled side-metric—and is visible per model as the corpus-reliance spectrum of Table 5.

(iii) **Why an item is not needed: *redundant* vs. *unusable*.** A small ablation-delta says the corpus did not move behavior, but not *why*, and the two reasons demand opposite fixes. We split a LEAKING item by its *with-corpus* performance: *redundant* (the learner is competent *with* the item present—it already knows the rule, so the item is dead weight and should be pruned) versus *unusable* (the learner fails the task *even with* the item present—the content has value but this model cannot act on it, so the fix is the model/prompting/retrieval, not the corpus). The delta alone conflates these; the with-corpus outcome (regret in the control domain, accuracy in QA) separates them. On the real-model sweep this is not hypothetical (Table 5): Claude Opus relies on every hazard in the suite while Llama-4-Maverick grounds on the override reaches *even with the rule present* (4/8 UNUSABLE)—*unusable*, not *redundant*.

Corpus sufficiency, and false sufficiency. Rolling the per-item verdicts to the corpus level yields a sufficiency judgement for a query set: *contributing* (every item relied upon), *partial*, *unusable*, or—the diagnostic ordinary RAG metrics cannot produce—*false sufficiency*: task accuracy is high yet necessity is ≈ 0 everywhere, so the corpus is not carrying the load and success rides on priors, fragile to distribution shift. The claim is explicitly scoped to the probed query set (sufficiency is always “for these queries”); the false-sufficiency signal (necessity ≈ 0) is precisely the measurable warning that a bounded-query sufficiency claim will not survive a shift.

Offline recovery of known ground truth. Before the real-model sweep, we confirm the verdicts recover truth we control. Against two reference mock learners whose truth is known, the instrument recovers it exactly: the corpus-bound mock yields ablation-delta 1.000 (regret-ablation 1981), follows the counterfactual, and abstains closed-book ($\Rightarrow$ CORPUS-BOUND); the leaking mock yields ablation-delta 0.000 (regret-ablation 0), ignores the counterfactual, and answers closed-book from priors ($\Rightarrow$ LEAKING).

Real-model discrimination (CS1.3). We run the necessity test as a *suite*: eight independent, alternating-bow charted hazards, each isolated in its own corpus, so per-model necessity is the *fraction of the suite relied upon* (resolution 1/8)—the fraction-valued reading of CS1.4 that supersedes an earlier single-hazard $\delta_{\mathcal{R}}$ scan (kept in the released provenance). Five production models run over AWS Bedrock—Claude Opus/Haiku/Sonnet 4.5, Amazon Nova-micro, and Llama-4-Maverick-17B—at a temp-0 point estimate plus a $K=5$ temp-0.7 ensemble (Table 5). Three findings.

(1) *A graded corpus-reliance spectrum.* The corpus-only hazard suite discriminates sharply where standard COLREG cannot: reliance falls cleanly from Claude Opus (relies on all 8) through Haiku (4), Sonnet (3), Nova-micro (2), to Llama-4-Maverick (0, which never reads the fact and grounds on the override reaches even with it present). This is the cross-model dynamic range that textbook COLREG—already parametric in every model—cannot produce, laid out as a spectrum rather than the near-binary split of the legacy single-hazard scan.

(2) *Read on necessity and regret-with, not the verdict.* We classify each hazard on *necessity* = $\mathcal{R}(\text{ablated}) - \mathcal{R}(\text{with-corpus})$ together with *regret-with*, not on the categorical leakage verdict. The verdict is a noisier three-signal majority vote that also demands following the inverted counterfactual and abstaining closed-book, so it *undercounts* a model that reads the fact (high necessity) but happens to answer closed-book from priors: an earlier verdict-based count wrongly showed Haiku at 0, where the necessity reading correctly credits it with 4. The continuous necessity and regret-with are the reliable signals; the verdict is retained only for the standard-COLREG strict-mode reading below.

(3) *Strict-mode faithfulness on standard COLREG.* Opus reads *standard* COLREG as CORPUS-BOUND, but the audit log shows this is a *strict-mode* effect, not a Rule-14 knowledge gap: when the steering rule is ablated Opus *abstains* (“no rules were provided...”, obeying the instruction to use only the provided rules) rather than falling back on its obvious parametric Rule-14 knowledge. Haiku instead falls back to Rule-14 starboard (LEAKING); Sonnet is split across samples (INCONCLUSIVE). Provenance: `experiments/provenance/cross-model/`.

Per-rule, not just per-corpus. To show the diagnosis attributes reliance to a *specific* rule rather than to corpus presence in general, we add a second, independent rule—Rule 15 crossing give-way—exercised on its own crossing scenarios (a rule only moves the metric on scenarios that invoke it, so each probe carries a matched subset). Ablating each rule produces a large regret signal on its own en-

Model	relied	unusable	redundant
Claude Opus 4.5	8	0	0
Claude Haiku 4.5	4	0	4
Claude Sonnet 4.5	3	2 [†]	3
Amazon Nova-micro	2 [†]	2 [†]	4
Llama-4-Maverick-17B	0	4	4

Table 5: Cross-model necessity over the eight-hazard corpus-only suite (five models, AWS Bedrock; CS1.3). Each hazard is isolated in its own corpus, so the three counts partition the suite of 8 and per-model reliance is a fraction (CS1.4’s fraction-valued fix, superseding an earlier single-hazard $\delta_{\mathcal{R}}$ scan). Classification uses *necessity* = $\mathcal{R}(\text{ablated}) - \mathcal{R}(\text{with-corpus})$ and *regret-with*: **relied** = necessity > 100 and regret-with < 10 (clears with the item, grounds without); **unusable** = regret-with ≥ 10 (grounds *with* the item present—present-but-unexploited); **redundant** = otherwise (clears both ways). The thresholds sit in wide empty gaps: the necessity signal is the clear $\leftrightarrow$ ground swing ($\sim 1400\text{--}1782$) against ≈ 0 noise, and regret-with ≥ 10 is a clean fail-with-corpus cut. Values are the temp-0 point estimate; the $K=5$ temp-0.7 ensemble has sd ≈ 0 except ([†]): Sonnet unusable 1.2 ± 0.7 , Nova-micro relied 0.2 ± 0.4 and unusable 2.2 ± 0.7 . Reliance is graded from Opus (8/8) to Maverick (0/8) where standard COLREG cannot discriminate. Provenance experiments/provenance/cross-model/.

266 counters: on `gemini-flash-latest` under the bound prompt, ablating Rule 14 gives $\delta_{\mathcal{R}}=1981$
267 (head-on) and ablating Rule 15 gives 1285 (crossing); on `gemini-flash-lite-latest`, 658
268 and 662 respectively. The two rules are localized on disjoint scenario subsets, so the instrument
269 reads reliance per rule.

270 **A real fact can be corpus-only (screen check, CS3.1).** A fair worry about the hazard arm is
271 that our danger is *fictional*, so “the model is corpus-bound” might just mean “the model cannot
272 know an invented fact.” We answer the narrower, real question this raises—*can a genuine, publicly-*
273 *charted fact be corpus-only for a frontier model?*—with a closed-book *screen*: we ask Claude Opus
274 to name each candidate danger unprompted and keep only the ones it cannot recall. Two obscure,
275 official-gazetteer-verified rocks—Elwha Rock (Cayou Channel, Puget Sound) and Fullastern Rock
276 (Adelaide Island, Antarctica)—survive the screen (Opus does not name them), while Whittle Rock
277 (False Bay) is *screened out* (Opus names it unprompted, so it is parametric, not corpus-only)—the
278 screen working as designed. We stress what this leg does and does not show. It validates the *screen*:
279 a real fact *can* be corpus-only for a current model, so the necessity phenomenon is not an artifact
280 of using invented dangers. It does *not* add an independent measurement of the necessity magnitude:
281 because the hazard’s identity never enters the objective J (only its rule text and location vary in
282 a fixed rig), any hazard Opus reads yields the same full-barrier $\delta_{\mathcal{R}} \approx 1998$ *by construction*—the
283 number is invariant to which danger it is, not a fresh confirmation. The screen is also imperfect:
284 a famous positive control, the Seven Stones reef, failed to screen out for the wrong reason (Opus
285 recalled adjacent dangers rather than its track), which we report rather than force. The load-bearing
286 evidence that necessity is *graded and calibrated* is the simulator-free fact-QA dose-response (Sec-
287 tion 6.1, CS3.2), where the metric is answer-accuracy over many independent facts and carries no
288 fixed-geometry confound.

289 **What in the corpus changes the answer: presence vs. retrieval.** The leakage test asks whether
290 the learner relies on the corpus *at all*; a deployed RAG tutor raises a finer question—of the
291 value curation adds, how much comes from the fact being *present* versus the retriever actually
292 *surfacing* it. Decomposing over in-corpus $\times$ retrieved gives three scorable conditions: NONE
293 (closed-book), RETRIEVED (the deployed dense retriever’s top- k), and ORACLE (the answer-bearing
294 chunk). Accuracy across them attributes the gain: $\text{acc}(\text{ORACLE}) - \text{acc}(\text{NONE})$ is the *content ceiling*,
295 $\text{acc}(\text{RETRIEVED}) - \text{acc}(\text{NONE})$ what the pipeline *delivers*, and their difference the *retrieval gap*. On
296 seven equipment-SOP fact-recall queries through a small local generator (Qwen2.5-1.5B), closed-
297 book is 0/7, ORACLE 7/7, and the deployed MiniLM pipeline 3/7—so of a 100% content ceiling
298 it delivers 43% and forfeits a 57% retrieval gap (generator-independent recall@5 4/7 localizes it).
299 The binding constraint is retrieval, not coverage or capability—distinct outcomes the binary leakage
300 verdict cannot separate. (One small model, seven queries: the method, not a benchmark. Harness:
301 experiments/retrieval-decomp.)

5 Case Study 2 — from blind to expert: reading quality and reasoning depth

Necessity is always measured against a *learner*, so it helps to name the cast. We arrange them on a *learner ladder* of increasing corpus use—**blind** (cannot perceive the danger; holds course and collides), **naive** (sighted but untrained; avoids, but generically—wrong side or over-bold), **implementer** (applies a corpus rule verbatim where the situation matches the corpus), and **reasoner** (combines the corpus with base rules and partial observation for novel or ambiguous cases)—with an off-ladder **parametric expert** that is competent *without* the corpus (it already knows the rule in its weights; necessity ≈ 0). Each rung-gap is a distinct corpus contribution and a distinct measurement. Case Study 1 (Section 4) lives on the source axis—*anything* vs. *parametric expert*, i.e. necessity itself. This section walks the depth ladder’s lower rungs: blind→naive is the *existence* gap, naive→implementer the response-*quality* gap (CS2.1), and implementer→reasoner the *generalization* gap (CS2.2).

Perception is a second source of the information the corpus can carry, which couples *what the corpus must supply* to *whether the learner can see*: an existence-corpus (“a danger is there”) is necessary only when the learner is *blind*, and a quality-corpus (“the right way to pass it”) only when the learner is *sighted* (perception already gives existence but not which-side or how-much). This is exactly why CS1 *hides* the hazard—so existence must come from the corpus and the learner must be blind—while CS2 *shows* it—so existence is free and the corpus must earn its keep on quality.

CS2.1 — decision quality (the middle band). The transfer objective of Section 3 is deliberately *bimodal*: a single hard barrier (collision/grounding, ≈ 2000) dominates the graded terms $\sim 1000\times$, so read directly it measures *existence* (did you avoid at all), not *quality* (did you avoid well). Reading quality off its *own* axis—the graded terms (ship-margin + compliance + route-deviation) *conditional on clearing*—separates three sailors the barrier alone conflates: **blind** collides (the barrier fires), **naive** clears but the wrong way (quality-regret ≈ 0.87 , the middle band), and **trained** clears lawfully (quality-regret ≈ 0.01 , the floor). A turn-angle sweep traces a smooth graded quality surface whose minimum sits at the trained maneuver, so the middle band is a genuine continuum, not a third discrete level. *Honest note*: this graded band comes from the smooth quality terms; a hard *second* hazard (a shoal on the wrong-side escape) yields only a thin feasibility shell rather than a graded surface, and that harder version is deferred to CS2.2’s overriding cases.

CS2.2 — reason vs. implement (the generalization axis). The deepest rung separates two *trained* learners that are identical on textbook cases and diverge only where a corpus *local rule* must override the reflex. We add a reasoning-aware objective in which a corpus-supplied local rule redefines the governing give-way side (COLREG Rules 2 and 9 subordinate the default steering rules to local conditions and good seamanship); when the local rule is absent the default starboard governs, so every prior scenario and result is unchanged. An *implementer* that looks the rule up and applies it verbatim and a *reasoner* that treats the corpus as premises then behave identically where the textbook rule already fits, and diverge only where the local rule overrides it. On a matched suite the two are indistinguishable (implementer $\equiv$ reasoner, $\Delta\text{regret } 0.00$); on a conflict suite built so the safe side *alternates* port/starboard—so no fixed reflex can pass—the **reasoner clears** 6/6 while the **lookup implementer clears only the 3** reaches where the local rule is redundant with Rule 14 and *grounds on the 3 it overrides*. The necessity of the local-rule corpus is thus 3/6, separable *only* on the overriding cases—the fact-vs-reasoning distinction (RAG-as-lookup vs. RAG-as-reasoning-substrate) made measurable. As in CS2.1 these are hand-built reference policies with known ground truth, so the run validates the *instrument*.

CS2.2 (live) — a real model reads as a reasoner. Putting a live model through the *same* B2 rig—necessity = $\mathcal{R}(\text{ablated}) - \mathcal{R}(\text{with-corpus})$, two conditions per reach, a temp-0 point estimate plus a $K=5$ temp-0.7 ensemble—makes the model the measured *subject*. **Gemini-2.5-flash** reads as a **reasoner**. On the three *override* reaches (deep water to port, where the Rule-14 starboard reflex grounds) it relied on the local rule in 5/5 samples, necessity 1782 ± 0 : with the corpus it alters to port and clears, and ablated it applies Rule 14 and grounds. On the three *redundant* reaches (where the local rule agrees with Rule 14) the necessity is scattered instead (relied 1–2/5, $\text{sd} \approx 800$)—the audit log shows the model *intermittently abstains* in the ablated arm (“the situation description contains contradictory information” → holds course → grounds → a spurious large necessity). *Scope note*: with only six draws per reach this is the *observed* behavior, not an abstention rate; the rea-

soner reading rests on the override reaches (5/5, zero variance), and the redundant-reach scatter is exactly what the ensemble’s variance control flags as noise rather than reliance. Provenance: experiments/provenance/reason-implement/gemini-2.5-flash.b2.jsonl.

6 Validating necessity at the weight level

The necessity/leakage test infers a fact about a model’s *weights* from a change in the *prompt*: ablate an item from the corpus, and if the governed outcome does not move, the model did not need it. The sharpest objection is that corpus *context* removal is not weight-level *knowledge* change, so a small necessity could be the no-context heuristic in disguise—or, conversely, a large necessity could reflect a fact the model happens not to know for reasons we did not control. We close this two ways, in increasing strength.

6.1 Validation by construction: the necessity dose-response

The holding this section establishes: *necessity is a non-confounded, calibrated measure*—teaching a fact into the weights drives it down as a dose-response—so a near-zero reading is evidence the model already knew the item, not a context-removal artifact. The decisive test is *known-groups*: build models along a gradient from fact-naïve to fact-knowing *by construction*, and show necessity falls monotonically. We teach a target fact into the weights (LoRA SFT) and sweep two independent knowledge gradients—the LoRA scale α (one adapter, evaluated at $\alpha \in [0, 1]$) and training checkpoints—so their agreement rules out a gradient-method artifact. We run this in two domains with *independent objectives*, the second of which uses no simulator at all.

Domain 1 — control-regret (a hidden hazard), CS1.7. Standard COLREG rules have little dynamic range for this test—every current frontier model already knows them (Section 4)—so we use a *corpus-only charted hazard*: a danger scored by the objective barrier but not shown in the prompt, knowable only from the corpus. Teaching it into a Qwen2.5-3B model collapses necessity from a full-barrier swing ($\mathcal{R}$ -necessity ≈ 667 , naïve) to ≈ 0.2 (taught), and the α and checkpoint sweeps *agree* on the transition. The curve is a *step*, not graded—the correct signature for a *single discrete fact*, which is learned as a threshold (the model either lacks the fact and grounds, or has it and clears)—so the hazard arm establishes the large-effect known-groups *endpoints*, and the graded curve is Domain 2’s job.

Domain 2 — simulator-free fictional-fact QA, CS3.2. We invent facts about fictional entities (research outposts and their attributes) that no model can have memorized, and score answers with a programmatic checker—*no simulator, no reference policy*. Because the facts are fictional, a competent model is necessarily corpus-bound at baseline, and because there are many *independent* facts, partial teaching learns a fraction of them, so the aggregate necessity is *graded*. Teaching the facts into a Qwen2.5-7B-Instruct model drives mean necessity smoothly and monotonically down the LoRA- α gradient: $1.00 \rightarrow 0.93 \rightarrow 0.29 \rightarrow 0.03$ (Table 6), with the verdict flipping CORPUS-BOUND $\rightarrow$ LEAKING as the facts are internalized. A training-checkpoint sweep reproduces the same curve ($1.00 \rightarrow 0.25 \rightarrow 0.03$)—the two gradients are built differently (scaling one converged adapter vs. genuine partial training), so their agreement rules out a gradient-method artifact. This is the clean dose-response, in a domain we did not build a simulator for—answering the objection that the measure is an artifact of one hand-built environment. (Necessity here is measured closed-book: the without-corpus condition permits parametric answering, so taught knowledge surfaces rather than being suppressed by a strict “answer only from the provided facts” instruction.)

6.2 Validation by unlearning: a *says* $\neq$ *does* dissociation (supporting, CS4.3)

The complementary direction removes a rule the model *does* know and asks whether necessity rises. This arm is secondary—it has dynamic range only on a rule with a real corpus-vs-prior gap, which the standard rules lack—but it yields an independently interesting result. We remove the alter-to-starboard knowledge from an open-weight model with a real unlearning method—SimNPO [Fan et al., 2024] as primary (reference-free, length-normalized), with NPO [Zhang et al., 2024], RMU [Li et al., 2024], and gradient ascent [Jang et al., 2022] as baselines—and score the base vs. unlearned model on the reference-policy instrument. A removal audit [Lynch et al., 2024] reports

Domain / objective	necessity across the knowledge gradient				
	naive	... step ...			taught
Hazard / $\mathcal{R}$ (Qwen-3B)	667				0.2
Fictional-fact QA / accuracy (Qwen-7B)	$\alpha=0$	0.25	0.5	0.75	$\alpha=1$
	1.00	0.93	0.29	0.03	0.03

Table 6: Necessity dose-response by construction. Teaching a fact into the weights drives the instrument’s necessity for that fact down. The hazard (control-regret objective) gives a large-effect *step*—the known-groups endpoints for one discrete fact, with α /checkpoint sweeps agreeing—while the simulator-free fictional-fact QA (answer-accuracy objective) gives the *graded* monotone curve over many independent facts. Two objectives, one measure. Harness: `experiments/unlearning` (`dose_response.py`), `src/engine/factqa`.

whether the knowledge is *gone* vs. merely *suppressed*; for a stable novice that resists benign re-learning we layer in a robust utility-preserving recipe [Singh et al., 2025]. A completed GPU run (gentle, guardrailed SimNPO on Qwen2.5-3B; config in Table 7) delivers not just the audit half but a first instrument-scoring result, whose lesson is sharper than the predicted 2×2 : **the standard free-text unlearning metrics overstate removal because they measure what the model says; the reference-policy instrument measures what the model *does*—its steering decision—and shows the decision is untouched.**

Primary result: a *says* $\neq$ *does* dissociation. Every metric that reads the model’s *words* registers forgetting: on the GPU run the forget-probe “starboard” rate falls $0.43 \rightarrow 0.27$, the forget-set NLL rises $7.4 \rightarrow 32.4$ (retain-set NLL, teacher-forced, even *improves* $4.3 \rightarrow 1.07$), and—most tellingly—the Rule-14 *citation* disappears, the model reattributing its maneuver to Rules 15/16/17. (A CPU run reproduces the likelihood collapse method-agnostically: forget NLL $4.6 \rightarrow 92$, direction-cue $5/6 \rightarrow 0/6$.) Yet the *decision* the instrument scores does not budge: the unlearned model’s completions *still turn* $+30^\circ$ to starboard (model damage flat at 0.10, retain-coherence 0.91)—the one quantity with real dynamic range here (the aggressive recipe below *does* move it, to port) stays put while every words-level metric registers forgetting. We separate this from the *verdict*: the 2×2 also does not move (both base and unlearned read LEAKING, ablation-delta 0.000), but that half carries *no* information, because Rule 14 already leaks at baseline, so ablating it is a null with no dynamic range for *any* weights (we make this explicit below). The dissociation rests on the heading, which *can* move and does not; the frozen verdict is a consequence of the baseline, not a measurement. A words-level metric reports a fact removed while the policy that governs behavior is entirely intact.

Suppressed, not removed (benign relearning). A gone-vs.-suppressed test [Hu et al., 2024, Deeb and Roger, 2024] confirms the reading from the other side. Twenty benign fine-tuning steps pull the forget-set NLL $32.4 \rightarrow 9.2$, back near its base value—the textbook *suppressed-not-removed* signature: the fact was never gone, only pushed down. Across the whole pipeline—base $\rightarrow$ unlearned $\rightarrow$ relearned—the one quantity that never moves is the decision the instrument measures: starboard, verdict LEAKING, at every stage. The likelihood rises and falls; the policy does not.

Contrast: aggressive unlearning *inverts* the rule. The gentleness is load-bearing. An earlier, aggressive recipe ($\text{lr } 1 \times 10^{-4}$, `retain_weight` 1.0, no guardrail) drove model damage $0.10 \rightarrow 0.33$ and turned the ship to *port*—the wrong way—on 6/8 prompts, moving the 2×2 to 6/8 port and verdict INCONCLUSIVE. Because the head-on turn is essentially binary, the default objective (push down “starboard”) simply relocated the mass onto “port,” *inverting* the safety rule rather than removing it—a caution that naive weight-level unlearning of a safety rule can yield *confident non-compliance*. The two recipes bracket the finding: gentle unlearning suppresses the *words* but not the *policy* (the instrument stays flat and so exposes the illusion), while aggressive unlearning moves the instrument—in the wrong direction. In both, the probe metrics track words and the instrument tracks the decision.

Why the 2×2 is flat. Rule 14 is already LEAKING at baseline—every current frontier model knows head-on $\rightarrow$ starboard parametrically (Section 4)—so ablating it changes nothing (ablation-delta 0), leaving no dynamic range for the corpus-bound cell to travel. The LEAKING verdicts are therefore

Metric	1.5B (CPU, SimNPO)		3B (GPU, gentle SimNPO)	
	base	unlearned	base	unlearned
Forget-target NLL ($\uparrow$)	4.6	92.0	7.4	32.4
Retain-set NLL, teacher-forced ($\downarrow$)	3.2	0.18	4.3	1.07
Forget-probe “starboard” rate ($\downarrow$)	5/6	0/6	0.43	0.27
Retain-coherence, free-gen ($\uparrow$)	—	—	1.00	0.91
Model damage (wrong+degen., $\downarrow$)	—	—	0.10	0.10
<i>Steering decision (instrument)</i>	—	—	<i>stbd</i>	<i>stbd</i>

Table 7: Gentle, guardrailed SimNPO unlearning of the alter-to-starboard rule, on a 1.5B model (CPU) and Qwen2.5-3B (GPU, `bfloat16`; $lr\ 5 \times 10^{-5}$, `retain_weight` 3.0, Rule-14 kept in the retain set). Every *words-level* metric registers forgetting—the forget-target likelihood collapses (NLL $\uparrow$), the “starboard” probe rate drops, and the Rule-14 citation disappears (model cites 15/16/17)—while retain utility is preserved (retain NLL improves; free-generation coherence 0.91) and there is no inversion (model damage flat at 0.10). Yet the *decision* the reference-policy instrument scores is untouched: the unlearned model still turns $+30^\circ$ to starboard, and the 2×2 stays verdict LEAKING, ablation-delta 0.000—a *says* $\neq$ does dissociation. Benign relearning (20 steps) pulls forget NLL $32.4 \rightarrow 9.2$, confirming the fact was *suppressed*, not removed; the decision holds throughout. Harness: `experiments/unlearning`.

446 *correct*; this is exactly why the primary weight-level validation is the construction dose-response
447 (Section 6.1), where a corpus-only fact *does* have range, and not this arm. A broader suite of corpus-
448 only probes across effect sizes (a medium-effect routing convention, a conflicting fact about another
449 vessel’s intention) is future work (Section 8).

450 7 Proxy learners as a pre-trial screen (application)

451 With the measure validated, its intended use is a pre-trial screen. We exercise the instrument with *two*
452 *complementary* controlled-competence proxy classes: (a) a parametric competence-vector learner,
453 and (b) a knowledge-suppressed / unlearned LLM learner (the arm of Section 6.2). Because both
454 are scored on the *same* $\mathcal{R}$ instrument, a method that fails to move the instrument in the expected
455 direction is a *no-go* before any human is recruited—exactly what the *says* $\neq$ *does* dissociation flags
456 for naive unlearning. This differs from prior proxy-learner work [Apartsin et al., 2026, Liu et al.,
457 2026b, Song et al., 2026, Fan et al., 2026, Lee et al., 2026] in scoring on an objective control-task
458 outcome rather than dialogue behavior, profile alignment, or LLM-judge helpfulness.

459 8 Limitations and scope

460 **Mechanism, not external validity.** Every result here is simulation-based mechanism evidence. We
461 make no learning-gains claim and pre-register the human cohort study as future work. **Proxy va-**
462 **lidity.** Simulated learners can be unfaithful—prompt-suppressed and unlearned models may be too
463 capable, may learn unrealistically fast, and need not reproduce authentic human misconceptions;
464 recent work cautions that student simulation is valid only under specific conditions [Yuan et al.,
465 2026] and documents systematic fidelity gaps between LLM “students” and real learners [Wu et al.,
466 2025, Srivatsa et al., 2025]. We therefore treat the two proxy classes as a mutual cross-check, not a
467 substitute for human data—which is exactly why the human cohort study is a pre-registered require-
468 ment rather than an optional extension. **Instrument sensitivity.** The protocol embeds weighting
469 choices (ship domain, objective); we show the ranking survives a $\pm 40\%$ reweighting ($\tau = 1.00$,
470 §3), but robustness beyond that range, and to the elliptical-domain *form* itself, is untested. **Small**
471 **evidence base — illustrative, not a benchmark.** The numbers here demonstrate that the *method*
472 behaves as designed; they are not a validity claim at scale. The discrimination result (Table 5) covers
473 five models across *three* provider families (Anthropic, Meta, Amazon) on an eight-hazard suite; an
474 earlier single-hazard scan (in the released provenance) additionally spanned eight models and found
475 two open reasoning models (`gpt-oss` 20B/120B) unscorable by the single-shot instrument—their
476 reasoning did not reliably terminate within the decode budget, so whether necessity is a property of
477 the (corpus, learner) pair awaits a variance-controlled rerun. The released harness now exercises
478 *three* rules across *two* metric axes—turn direction (Rule 14 head-on, Rule 15 crossing) and, on a

restricted-visibility subset, speed (Rule 6/19 safe speed, scored through the decision’s speed rather than its heading)—and the unlearning arm turns on a single rule and model family. A persuasive claim still needs more models per family and many more rules/encounters—which is why we release the harness (`experiment.sh`, and a Bedrock provider $\times$ size sweep) built to run exactly that, and report these as worked examples rather than a leaderboard. **Single-rule decomposability may not generalize.** KC $\rightarrow$ single-metric identifiability and clean per-rule ablation are properties we *design into* a governed corpus and *verify* per instrument (Table 3); they are not guaranteed for an arbitrary corpus, whose rules may be entangled, redundant, or jointly govern one outcome. Where a single rule cannot be cleanly isolated, the localization degrades to a coarser attribution and the leakage test to a group-level one; how far clean decomposition extends beyond corpora we constructed is open. **The unlearning arm is a single model at one scale.** The *says $\neq$ does* result (Section 6.2) is informative—a gentle recipe suppresses the *words* while the scored *decision* is untouched, an aggressive one *inverts* the rule—but it is one 3B model, and seed/method variance is outstanding. It is a *supporting* result: a clean necessity transition is structurally out of reach for any pretrained COLREG rule (all leak), which is exactly why the primary weight-level validation is the *construction* dose-response (Section 6.1), where we build the knowledge into the weights and observe necessity fall—already run in two domains, though on single models per domain; more seeds and sizes (and a 7B hazard replication in `bfloat16`, not 4-bit, since quantization noise would confound the effect) remain breadth. **Sufficiency is scoped to a query set.** Every sufficiency verdict is relative to the probed queries; we mitigate rather than escape this by always reporting the query count and by treating the false-sufficiency signal as the measurable warning of fragility to shift, but a corpus adequate for one query distribution may be insufficient for another.

Future work. The pre-registered human cohort study is the external-validity test this in-silico screen is built to de-risk. Method-side, we plan to scale the cross-model discrimination (more providers and rule sets; the harness already runs class $\times$ size sweeps) and to broaden the weight-level battery from one discrete hazard to a *suite* of corpus-only probes across effect sizes—a medium-effect routing convention and a conflicting fact about another vessel’s intention, building on the real charted-danger leg already shown (Section 4)—so necessity is validated as a distribution across probe kinds, not a single test.

9 Conclusion

We have described an in-silico instrument that measures whether a learner *needs* a corpus, and whether a corpus is *sufficient*, before human-subject effort. The central contribution is a *necessity* measure made non-confounded by *construction*: teaching a fact into a model’s weights drives the instrument’s necessity for that fact down as a dose-response, which we demonstrate in two domains with independent objectives—a control-regret simulator (large-effect known-groups endpoints) and a *simulator-free* fictional-fact QA task (the graded monotone curve, $1.00 \rightarrow 0.03$). On that calibrated measure we build a corpus-value audit—ranking items by necessity, localizing a missing item, and splitting the unrelayed-upon into *redundant* versus *unusable*—and a corpus-sufficiency verdict that surfaces *false sufficiency*. Distinct from faithfulness and context-attribution, we measure need against a weight-level baseline they lack. Supporting breadth—a reference-policy transfer instrument with KC $\rightarrow$ single-metric identifiability, a learner ladder that reads corpus contribution beyond existence (decision quality and reasoning depth), cross-model discrimination, a second control domain, and a *says $\neq$ does* unlearning dissociation—rounds out the picture. The claim is testable and narrow; the domains are worked examples, the evidence base is deliberately small, and the measurement method is the point.

Reproducibility. The simulator, the answer-checker QA domain, the diagnosis and sufficiency harness, the construction dose-response, and the open-weight unlearning scaffold will be released, along with a deterministic no-key dry-run that recovers the mock ground truth and one-tap GPU notebooks for the dose-response. The whole offline backbone regenerates in one deterministic, LLM-free command that checks each headline result against its claimed value (CS4.4). (Code is withheld here to preserve double-blind review.)

References

Anonymous. Quantitative COLREG compliance evaluation for autonomous ships. *Ocean Engineering*, 2024a.

532 Anonymous. Control knowledge tracing: A control-theoretic view of knowledge dynamics. In *Conf. on AI in*
533 *Education (CAEAI)*, 2024b.

534 Alexander Apartsin, Omri Sason, and Yehudit Aperstein. Toward a benchmark for controllable simulation of
535 imperfect students with large language models, 2026.

536 Susan Bull and Judy Kay. Student models that invite the learner in: The SMILI open learner modelling frame-
537 work. *International Journal of Artificial Intelligence in Education*, 17(2):89–120, 2007.

538 Benjamin Cohen-Wang, Andrew Ilyas, Jonathan Wu, and Aleksander Madry. Contextcite: Attributing model
539 generation to context. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

540 Albert T. Corbett and John R. Anderson. Knowledge tracing: Modeling the acquisition of procedural knowl-
541 edge. *User Modeling and User-Adapted Interaction*, 4(4):253–278, 1994.

542 Aghyad Deeb and Fabien Roger. Do unlearning methods remove information from language model weights?,
543 2024.

544 Shahul Es, Jithin James, Luis Espinosa-Anke, and Steven Schockaert. RAGAS: Automated evaluation of
545 retrieval augmented generation, 2023.

546 Chongyu Fan, Jiancheng Liu, Licong Lin, Jinghan Jia, Ruiqi Zhang, Song Mei, and Sijia Liu. Simplicity
547 prevails: Rethinking negative preference optimization for llm unlearning, 2024.

548 Shuyi Fan, Boyuan Deng, Mengyu Xu, Jiale Liu, Hongyang Zhang, Qiaoxin Yang, and Chongyang Gao. Re-
549 thinking llm-judged helpfulness as a pedagogy signal: A pre-registered audit across tutor models, 2026.

550 Paolo Fiorini and Zvi Shiller. Motion planning in dynamic environments using velocity obstacles. *International*
551 *Journal of Robotics Research*, 17(7):760–772, 1998.

552 Weibo Gao, Qi Liu, Linan Yue, Fangzhou Yao, Rui Lv, Zheng Zhang, Hao Wang, and Zhenya Huang.
553 Agent4edu: Generating learner response data by generative agents for intelligent education systems, 2025.

554 Shengyuan Hu, Yiwei Fu, Zhiwei Steven Wu, and Virginia Smith. Unlearning or obfuscating? joggling the
555 memory of unlearned llms via benign relearning, 2024.

556 International Maritime Organization. Convention on the international regulations for preventing collisions at
557 sea (COLREGs), 1972.

558 Siddharth Jain and Venkat Narayan Vedam. Cue-r: Beyond the final answer in retrieval-augmented generation,
559 2026.

560 Joel Jang, Dongkeun Yoon, Sohee Yang, Sungmin Cha, Moontae Lee, Lajanugen Logeswaran, and Minjoon
561 Seo. Knowledge unlearning for mitigating privacy risks in language models, 2022.

562 Tor A. Johansen, Tristan Perez, and Andrea Cristofaro. Ship collision avoidance and COLREGS compliance
563 using simulation-based control behavior selection with predictive hazard assessment. *IEEE Transactions on*
564 *Intelligent Transportation Systems*, 17(12):3407–3422, 2016.

565 Kenneth R. Koedinger, Albert T. Corbett, and Charles Perfetti. The Knowledge-Learning-Instruction frame-
566 work: Bridging the science-practice chasm to enhance robust student learning. *Cognitive Science*, 36(5):
567 757–798, 2012.

568 Vishakha Lall and Yisi Liu. Ai meets maritime training: Precision analytics for enhanced safety and perfor-
569 mance, 2025.

570 Unggi Lee, Sookbun Lee, Yeil Jeong, Eunjoo Lee, Minchul Shin, and Hoilym Kwon. Educlaw-bench: A
571 long-horizon benchmark for pedagogical llm agents with simulated learners, 2026.

572 Nathaniel Li et al. The wmdp benchmark: Measuring and reducing malicious use with unlearning, 2024.

573 Jiayi Liu, Jiaxing Zhang, Bowen Jin, and Jennifer Neville. Generating leakage-free benchmarks for robust rag
574 evaluation, 2026a.

575 Ruochen Liu, Zhiyuan Wen, Hao Yan, Jun Yin, Senzhang Wang, and Jiannong Cao. Ps²: Parameterized control
576 for fine-grained student proficiency simulation, 2026b.

577 Xinyi Lu and Xu Wang. Generative students: Using llm-simulated student profiles to support question item
578 evaluation, 2024.

579 Aengus Lynch, Phillip Guo, Aidan Ewart, Stephen Casper, and Dylan Hadfield-Menell. Eight methods to
580 evaluate robust unlearning in llms, 2024.

581 Barak Or. Quantifying prior dominance in rag systems, 2026.

582 Radek Pelánek. Applications of the Elo rating system in adaptive educational systems. *Computers & Education*,
583 98:169–179, 2016.

584 Naman Deep Singh, Maximilian Müller, Francesco Croce, and Matthias Hein. Unlearning that lasts: Utility-
585 preserving, robust, and almost irreversible forgetting in llms, 2025.

586 Jiajia Song, Zhihan Guo, and Jionghao Lin. Simulating novice students using machine unlearning and relearn-
587 ing in large language models, 2026.

588 KV Aditya Srivatsa, Kaushal Kumar Maurya, and Ekaterina Kochmar. Can llms reliably simulate real students’
589 abilities in mathematics and reading comprehension?, 2025.

590 Adrienne Tuynman and Ronald Ortner. Transfer in reinforcement learning via regret bounds for learning agents,
591 2022.

592 K. W. Van Dongen et al. Construct validity of the LapSim: Can the LapSim virtual reality simulator distinguish
593 between novices and experts? *Surgical Endoscopy*, 21(8):1413–1417, 2007.

594 Lev S. Vygotsky. *Mind in Society: The Development of Higher Psychological Processes*. Harvard University
595 Press, 1978.

596 Tao Wu, Jingyuan Chen, Wang Lin, Mengze Li, Yumeng Zhu, Ang Li, Kun Kuang, and Fei Wu. Embracing
597 imperfection: Simulating students with diverse cognitive levels using llm-based agents, 2025.

598 Rongwu Xu, Zehan Qi, Zhijiang Guo, Cunxiang Wang, Hongru Wang, Yue Zhang, and Wei Xu. Knowledge
599 conflicts for LLMs: A survey. In *Proceedings of EMNLP*, 2024.

600 Zhihao Yuan, Yunze Xiao, Ming Li, Weihao Xuan, Richard Tong, Mona Diab, and Tom Mitchell. Towards
601 valid student simulation with large language models, 2026.

602 Qinggang Zhang, Zhishang Xiang, Yilin Xiao, Le Wang, Junhui Li, Xinrun Wang, and Jinsong Su. Faithfulrag:
603 Fact-level conflict modeling for context-faithful retrieval-augmented generation, 2025.

604 Ruiqi Zhang, Licong Lin, Yu Bai, and Song Mei. Negative preference optimization: From catastrophic collapse
605 to effective unlearning, 2024.

606 Zinn et al. Assessing flight competency from deviation to an optimal trajectory. *Cognition, Technology & Work*,
607 2023.